

Question Paper

Exam Date & Time: 27-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Analog Electronic Circuits [ECM_2121]

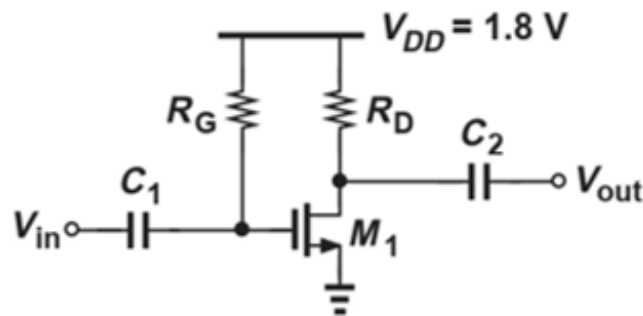
Marks: 50

Duration: 180 mins.

PART-A

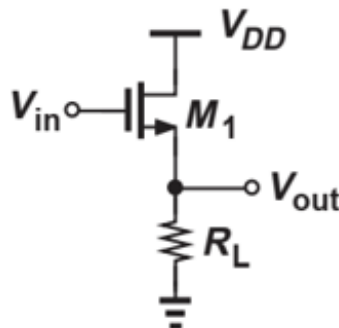
Answer all the questions.

- 1) Design the amplifier shown in the figure below, for a voltage gain of -5 and an output impedance of $1\text{K}\Omega$. Bias the transistor so that it operates 100 mV away from the triode region. Assume the capacitors are very large, $R_G = 10\text{K}\Omega$, $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $\lambda=0$, and $V_{TH}=0.4\text{V}$. (4)
- A)



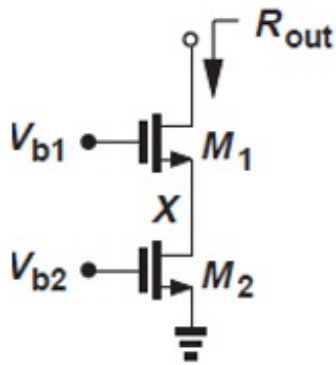
Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Using a small signal model, calculate the voltage gain for the circuit shown below. Consider $\lambda \neq 0$. (3)



Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) Design an NMOS cascode for an output impedance of $500 \text{ k}\Omega$ and a current of 0.5 mA . Assume $M1$ and $M2$ are identical, $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$ and $\lambda = 0.1\text{V}^{-1}$. (3)

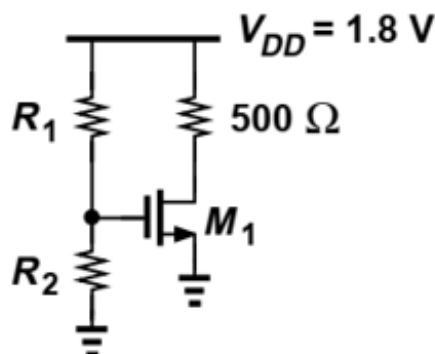


Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) Design an NMOS differential pair for a voltage gain of 5 and a power budget of 2 mW subject to the condition that the stage following the differential pair requires an input CM (common mode) level of at least 1.6 V. Assume $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $\lambda = 0$, and $V_{DD} = 1.8\text{V}$. (4)
- A)

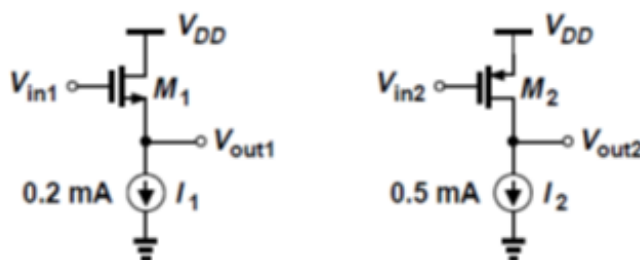
Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Assume $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $\lambda = 0$, and $V_{TH} = 0.4\text{V}$. For a drain current of 1 mA. If $W/L = 20/0.18$. For the circuit shown, compute R1 and R2 such that the input impedance is at least $20\text{k}\Omega$. (3)



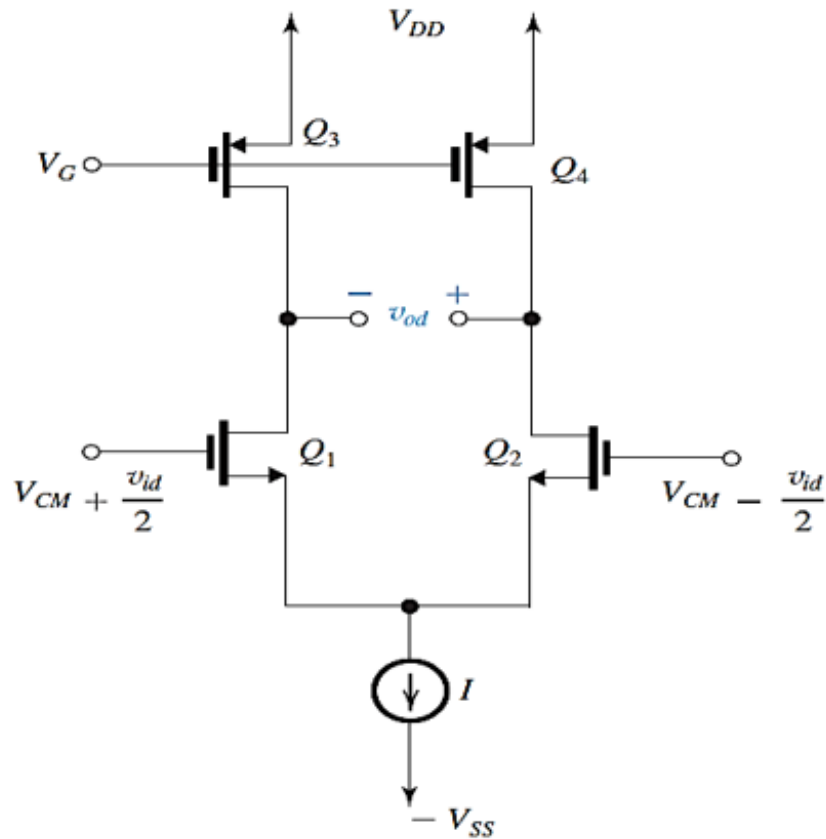
Course Outcome: "CO2" / Cognitive Level: "Application"

- C) An integrated circuit employs the source follower and the common-source stage for the circuit shown. Design a current mirror that produces I_1 and I_2 from a 0.3-mA reference. (3)



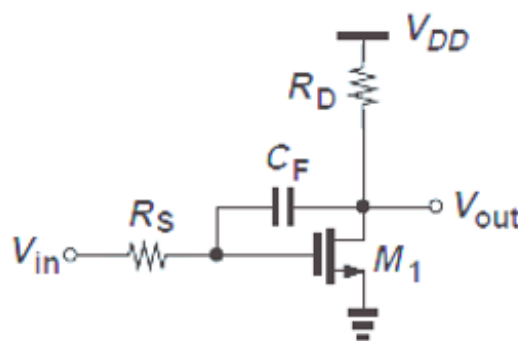
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) The differential amplifier of Fig. is fabricated in a 0.18- μm CMOS technology for which $\mu_n C_{ox} = 4\mu_p C_{ox} = 400 \mu\text{A}/\text{V}^2$, $V_t = 0.5 \text{ V}$, and $V_A = 10 \text{ V}/\mu\text{m}$. If the bias current $I = 200 \mu\text{A}$ and all transistors have a channel length twice the minimum and are operating at $V_{ov} = 0.2 \text{ V}$, find W/L for each of Q_1 , Q_2 , Q_3 , and Q_4 , and determine the differential voltage gain A_d . (4)
- A)



Course Outcome: "CO2" / Cognitive Level: "Application"

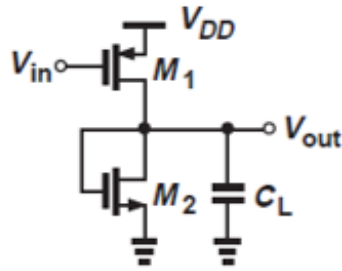
- B) Estimate the pole of the circuit shown. Assume $\lambda = 0$. Calculate C_{in} if $g_m = (200 \Omega)^{-1}$, $R_D = 2.5 \text{ k}\Omega$, and $C_F = 70 \text{ fF}$. (3)



Course Outcome: "CO3" / Cognitive Level: "Application"

- C) (3)

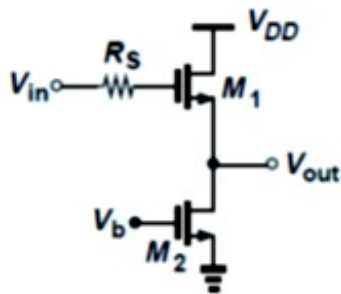
Compute the poles of the circuit shown in circuit. Assume $\lambda \neq 0$.



Course Outcome: "CO3" / Cognitive Level: "Application"

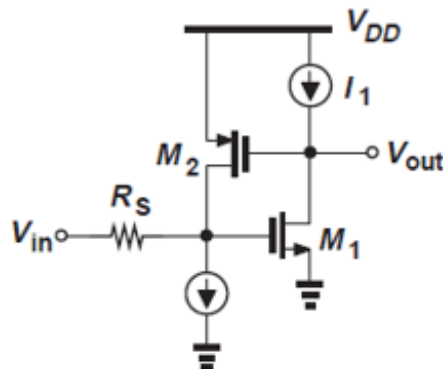
- 4) Identify all the capacitance and reduce the source follower circuit shown to compute the poles. Assume $\lambda=0$. (4)

A)



Course Outcome: "CO3" / Cognitive Level: "Application"

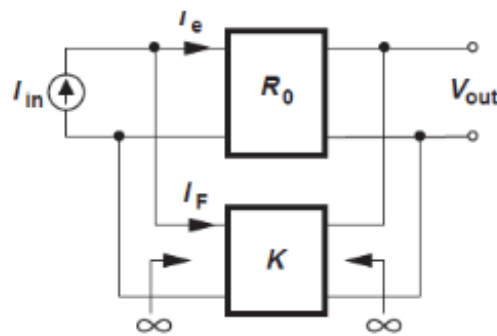
- B) Determine the polarity of the feedback of the circuit and justify the same. (3)



Course Outcome: "CO4" / Cognitive Level: "Understand"

- C) (3)

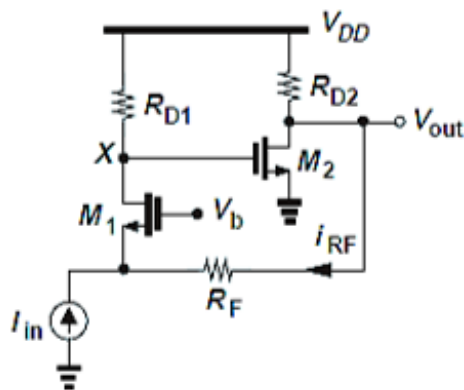
Identify the feedback topology for the circuit shown and obtain the expression for closed loop gain and I/O impedances.



Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Determine the closed-loop gain and I/O impedances of the feedback circuit shown. (4)

A)



Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Obtain the efficiency of Class - B Push Pull power amplifier with a neat circuit diagram. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) Explain the working principle of the Colpitt's Oscillator with a neat circuit diagram. Also, when $C_1 =$ (3)
 $C_2 = 0.002\mu\text{F}$ and $L = 100\mu\text{H}$, determine the oscillation frequency.

Course Outcome: "CO4" / Cognitive Level: "Understand"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECM_2121:Analog Electronic Circuits

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	21
3	3	CO 3	10
4	4	CO 4	13
5	5	CO 5	3
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECM_2121:Analog Electronic Circuits

#	Level	No of Questions	Marks (%)
---	-------	-----------------	-----------

1	Remember	0	0(0%)
2	Understand	5	15(30%)
3	Application	8	27(54%)
4	Analysis	2	8(16%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)